

Bounded Asymmetry Profiles for Comparing Protein Disorder Distributions Across Taxa: A Methodological Report with a Negative Concordance Result for an AlphaFold-Derived Proxy

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Abstract: Comparisons of biological feature distributions are usually reduced to a single test statistic, which establishes *whether* two populations differ but not *where* along the feature axis the difference lies. We define a bounded local asymmetry profile $S(x)$ together with three scalar summaries, and evaluate it on experimentally curated intrinsic disorder annotations for 1,766 proteins from four taxa (DisProt release 2026_06: 1,339 human, 223 *Saccharomyces cerevisiae*, 146 *Escherichia coli* K12, 58 *Caenorhabditis elegans*). Two findings are reported. First, the profile behaves as intended as a descriptive instrument: for the human-*E. coli* comparison of curated disorder fraction it resolves localised structure ($\bar{A} = 0.143$, $C = 0.57$) in a comparison that global tests read as null (Kolmogorov-Smirnov $p = 0.39$; Mann-Whitney $p = 0.85$). Second, and limiting, the accompanying AlphaFold-derived very-low-confidence fraction is a poor *protein-level* surrogate for curated disorder content in every taxon examined (Spearman $\rho = 0.07$ – 0.24), and its offset relative to curated disorder differs significantly between taxa (median signed gap $+0.001$ human versus -0.083 *E. coli*; Mann-Whitney $p < 10^{-4}$). We interpret this as a threshold-and-calibration effect rather than a contradiction of the established residue-level performance of pLDDT-based disorder prediction, and conclude that the very-low-confidence fraction should not be used as a taxon-neutral protein-level disorder feature without recalibration. We also document that the scalar magnitude index $\bar{A}$ is estimator-dependent: it inflates with histogram bin count because empty bins drive $|S| \rightarrow 1$, and the signed index changes sign in 5 of 12 comparisons under histogram estimation while remaining stable across all kernel bandwidths and regularisation constants tested. Kernel estimation with a reported bandwidth is therefore required for the framework to be reproducible.

Keywords: intrinsically disordered proteins; DisProt; AlphaFold; pLDDT; distributional comparison; cross-taxon calibration; exploratory methodology

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1. Introduction

Intrinsically disordered proteins (IDPs) and intrinsically disordered regions (IDRs) carry out molecular recognition, signalling and regulatory functions without adopting a single stable fold, and they are now recognised as a pervasive rather than exceptional feature of proteomes [1, 2]. Their prevalence is markedly uneven across the tree of life. Genome-scale predictions with DISOPRED2 placed long (>30 -residue) disordered segments in 2.0% of archaeal, 4.2% of eubacterial and 33.0% of eukaryotic proteins [3], and proteome-level analyses of *E. coli* and *S. cerevisiae* reached compatible conclusions [4]. Taxonomic differences in disorder content are therefore an established biological expectation, not a hypothesis this report needs to create.

Comparative studies of disorder across organisms typically summarise each population by a mean or median disorder fraction and test for a difference. Such summaries answer whether two populations differ but discard the *location* of the difference along the feature axis, so localised enrichment can be invisible in an aggregate that is close to null. This is a general limitation of scalar two-sample testing rather than a limitation specific to disorder biology, and

it motivates representations that retain positional information over the feature domain.

A second motivation is the recent availability of structure-prediction confidence as a cheap, proteome-scale disorder signal. AlphaFold2 [5] and the AlphaFold Protein Structure Database [6, 7] supply per-residue predicted Local Distance Difference Test (pLDDT) scores [8] for essentially the whole of UniProt. Because AlphaFold renders regions without stable structure as low-confidence extended ribbons, $1 - \text{pLDDT}$ functions directly as a residue-level disorder score. This has been validated repeatedly: Piovesan et al. [9] showed that pLDDT-based prediction is competitive on the CAID DisProt benchmarks, and Wilson et al. [10] found that pLDDT provides a good residue-wise measure of disorder, in contrast to naive secondary-structure-based assignment. The DisProt resource itself reports a strong relationship between its curated annotations and pLDDT-derived disorder [11]. Successive rounds of the Critical Assessment of protein Intrinsic Disorder (CAID) have quantified how such methods compare with sequence-based predictors [12, 13, 14].

Two caveats in that literature are directly relevant here. First, the established performance claims are *residue-level*

discrimination statistics, and the optimal pLDDT cutoff for disorder detection is not the standard “very low confidence” band boundary: Piovesan et al. [9] report an optimal classification threshold corresponding to pLDDT < 68.8, well above the pLDDT < 50 boundary of the very-low band. Second, evaluations that target protein-level *disorder content* rather than per-residue labels report weaker and more variable performance [15], and broader reviews have emphasised that AlphaFold confidence carries known pitfalls outside rigid globular proteins [16, 17, 18]. Whether pLDDT-derived features are calibrated *comparably across taxa* appears not to have been tested directly.

This report has a narrow and explicitly methodological aim. We define a bounded local asymmetry profile over a continuous feature coordinate, derive three scalar summaries from it, and ask two questions of curated disorder data: does the representation add interpretable information beyond global tests, and is an AlphaFold-derived confidence feature comparable enough across taxa to be used as a disorder proxy in later work?

The second question is prior to any biological application. If a structure-prediction confidence feature is calibrated differently in different taxa, then cross-taxon contrasts computed from it will confound calibration with biology. We therefore treat the concordance test as the gating result and report it even though it is largely negative.

2. Materials and Methods

2.1. Definition of the bounded asymmetry profile

Let $p_A(x)$ and $p_B(x)$ be normalised density estimates for two populations over a feature coordinate $x \in \Omega$. Define the bounded local asymmetry profile

$$S(x) = \frac{p_A(x) - p_B(x)}{p_A(x) + p_B(x) + 2\varepsilon}, \quad (1)$$

with regularisation constant $\varepsilon > 0$. Because $|p_A - p_B| \leq p_A + p_B$, Eq. (1) satisfies $-1 < S(x) < 1$ pointwise; it is antisymmetric under exchange of populations, $S_{A,B} = -S_{B,A}$; and $S(x) = 0$ exactly where the two density estimates agree. Positive values indicate local enrichment of A . The construction is a normalised difference of densities and is related in form to symmetric relative-density contrasts used elsewhere in applied statistics; the contribution claimed here is not novelty of the ratio but the pairing of a bounded profile with the three summaries below and an explicit account of its estimator dependence.

2.2. Scalar summaries

For a domain of length $|\Omega|$ we define the normalised asymmetry magnitude, signed balance index and directional consistency index

$$\bar{A} = \frac{1}{|\Omega|} \int_{\Omega} |S| dx, \quad \bar{B} = \frac{1}{|\Omega|} \int_{\Omega} S dx, \quad (2)$$

$$\mathcal{C} = \frac{|\bar{B}|}{\bar{A}} \quad (\bar{A} > 0), \quad (3)$$

with $\mathcal{C} := 0$ when $\bar{A} = 0$. From $-1 < S < 1$ and the triangle inequality,

$$0 \leq \bar{A} < 1, \quad -1 < \bar{B} < 1, \quad |\bar{B}| \leq \bar{A}, \quad 0 \leq \mathcal{C} \leq 1, \quad (4)$$

with $\bar{A} = 0$ iff $S \equiv 0$ almost everywhere, and $\mathcal{C} = 1$ iff S has constant sign almost everywhere (equality in the triangle inequality). The three quantities separate three questions that are easily conflated: how much do the distributions differ, which population is favoured on balance, and is that direction maintained across the domain. In this study $\Omega = [0, 1]$, so normalised and unnormalised indices coincide.

2.2.1. Scope and a caveat on ε

The construction requires only two normalised densities over a common coordinate and is otherwise independent of the application. One caveat follows directly from Eq. (1): ε carries the units of a density, so S is not invariant under reparameterisation of x unless ε is rescaled with it. Comparisons of $\bar{A}$ across different feature coordinates, or across density estimators with different peak magnitudes, are therefore not meaningful. Section 3.4 shows this is not a purely formal concern.

2.3. Source data

Four per-organism JSON exports of DisProt [11, 19] release 2026_06 (the “with ambiguous evidences” variant) were used. The files are mutually disjoint: 1,766 records, 1,766 unique UniProt accessions [20] and 1,766 unique DisProt identifiers, so no protein is double-counted. Composition, protein lengths and integrity checks are given in Table 1 and Fig. 1. Declared protein length matched the deposited sequence length for all 1,766 records, no accession or region boundary was missing, and no region extended past the protein terminus.

Two provenance limitations were identified during the audit and are carried forward. (i) For 545 proteins the exported `regions` array contains fewer entries than the record’s own `regions_counter` field (7,535 exported regions against a declared 9,610), so the export is a filtered view of the curated annotation set; the discrepancy could not be resolved against the live database (Section 2.4). (ii) The record-level `disorder_content` field reproduces, for all 1,766 proteins, the residue fraction covered by structural-state consensus segments of type `D` or `S`, i.e. it includes ambiguous-evidence residues. We therefore re-computed disorder fraction from type-`D` segments only,

$$D = \frac{\#\{\text{residues in type-D structural-state consensus}\}}{\text{protein length}} \in [0, 1], \quad (5)$$

and use Eq. (5) throughout. Only 4 proteins carry type-S segments, so the choice is consequential for provenance rather than for the numbers.

2.4. The AlphaFold feature, and what it is not

Each DisProt record carries a precomputed scalar field, `alphafold_very_low_content`, which we denote V and treat as the fraction of residues assigned very low AlphaFold confidence ($\text{pLDDT} < 50$ under the standard confidence bands of AlphaFold DB [5, 6]). This interpretation follows the field name and the published confidence-band convention, but it was **not verified against AlphaFold DB**, because every required database was unreachable: requests to the AlphaFold DB prediction API and confidence-JSON endpoints, the UniProt REST API, MobiDB [21] and the DisProt API all returned HTTP 403 with proxy reason `host_not_allowed` (logged in `alphafold_retrieval_manifest.csv`).

Three consequences must be stated plainly. No residue-level pLDDT values were retrieved, so no residue-level analysis was performed: no per-residue sensitivity, specificity, ROC or Matthews correlation, and no comparison of median pLDDT inside versus outside annotated regions. The exact definition and AlphaFold model version behind V are unverified; this matters because AlphaFold DB has been rebuilt across releases [7, 22] and because AlphaFold3 produces pLDDT distributions in IDRs that differ significantly from AlphaFold2 [23, 24]. Finally, V is absent for 103 proteins (92 human, 4 fungal, 3 bacterial, 4 nematode); only 18 of these exceed 2,700 residues, so model-length limits do not account for most of the missingness, and the cause remains unresolved. Those proteins are listed in `failed_accessions.csv` and excluded pairwise.

2.5. Estimation and inference

Densities were estimated on a 512-point grid over $[0, 1]$ by Gaussian kernel density estimation with Scott’s rule [26], renormalised to unit integral on the closed domain, with $\varepsilon = 10^{-3}$ unless stated. Renormalisation rather than reflection was used so that one estimator applies to every group; a histogram estimator is reported alongside as a check. Integrals in Eq. (3) were evaluated by the trapezoidal rule.

The protein was the resampling unit throughout: confidence intervals for $\bar{A}$, $\bar{B}$ and $\bar{C}$ come from 2,000 protein-level bootstrap replicates [27], drawn independently within each group. Distributional differences were additionally assessed by two-sample Kolmogorov–Smirnov and Mann–Whitney [28] tests with the rank-biserial correlation [29] as effect size. Concordance between D and V was measured by Spearman ρ with bootstrap intervals, and the within-protein signed gap $V - D$ was compared across taxa. All computations used SciPy [30]. Reported p -values are not corrected for multiplicity; with 12 pairwise comparisons per feature, values near 0.05 should be read as descriptive.

3. Results and Discussion

3.1. The profile resolves structure that global tests miss

Pairwise indices are given in Table 2, profiles in Fig. 2, and bootstrap intervals in Fig. 3.

The clearest illustration of what the representation adds is human versus *E. coli* on curated disorder. Global tests are null (KS $p = 0.39$; Mann–Whitney $p = 0.85$; rank-biserial -0.009) and the medians are close (0.124 versus 0.114), yet the profile is not flat: $\bar{A} = 0.143$ with $\bar{C} = 0.57$ and two zero crossings, indicating regions of opposing local enrichment that cancel in aggregate. The bootstrap interval for the signed index includes zero ($\bar{B} = 0.081$, 95% CI $[-0.036, 0.250]$), so the *direction* of that structure is not established at this sample size; what the profile establishes is that a null global test does not imply locally similar densities.

It is worth noting that a null human–*E. coli* contrast in curated disorder is itself a product of ascertainment rather than evidence against the prokaryote–eukaryote gap reported from proteome-wide prediction [3, 4]. DisProt *E. coli* entries exist because their disorder was investigated, which selects against the ordered majority of that proteome.

Two comparisons of curated disorder are directionally resolved. Fungal proteins are enriched relative to human ($\bar{B} = -0.125$, CI $[-0.178, -0.048]$; KS $p = 7 \times 10^{-4}$) and relative to bacterial proteins ($\bar{B} = 0.196$, CI $[0.077, 0.328]$; KS $p = 3 \times 10^{-3}$), consistent with the median ordering in Table 1. The nematode group ($n = 58$) yields intervals spanning zero in every comparison and supports no directional claim.

3.2. The AlphaFold proxy is weakly concordant and taxon-dependent

This is the gating result. Within-protein concordance between the curated disorder fraction and the AlphaFold very-low fraction is weak in every group (Fig. 4): Spearman $\rho = 0.07$ (95% CI $[0.01, 0.13]$, $n = 1246$) in human, 0.24 $[0.09, 0.37]$ in fungal, 0.18 $[0.01, 0.34]$ in bacterial, and 0.18 $[-0.11, 0.47]$ in nematode. The human interval excludes zero only because n is large; the point estimate corresponds to under 1% of shared rank variance. No group shows concordance that would justify substituting V for D at the protein level.

The offset is also taxon-dependent. The median within-protein gap $V - D$ runs from +0.001 (human) and +0.020 (nematode) to -0.041 (fungal) and -0.083 (bacterial), and the gap distributions differ significantly for human versus fungal, human versus bacterial, fungal versus bacterial, fungal versus nematode and bacterial versus nematode (Mann–Whitney $p \leq 5 \times 10^{-3}$; only human versus nematode is null, $p = 0.56$). The bacterial case is extreme: median $V = 0.028$ against a median curated disorder fraction of 0.114, with V exactly zero for 19.6% of *E. coli* proteins versus 2.6% of human proteins. Accordingly the

Table 1. Curated sample composition and protein-level feature summaries. D is the curated disorder fraction of Eq. (5); V is the AlphaFold very-low-confidence fraction as deposited in the export. “no V ” counts proteins for which the export carries no AlphaFold value.

Group (organism)	n	median		median		median	
		length	no V	D	IQR D	V	$V - D$
Human (<i>H. sapiens</i>)	1339	480	92	0.124	0.240	0.176	+0.001
Fungal (<i>S. cerevisiae</i>)	223	401	4	0.180	0.318	0.143	−0.041
Bacterial (<i>E. coli</i> K12)	146	267	3	0.114	0.182	0.028	−0.083
Nematode (<i>C. elegans</i>)	58	543	4	0.096	0.166	0.143	+0.020

Table 2. Bounded asymmetry indices and global tests. Positive $\bar{B}$ indicates net enrichment of group A . CIs are 95% protein-level bootstrap; z marks intervals excluding zero. n_0 is the number of zero crossings of S . Feature D = curated disorder fraction, V = AlphaFold very-low fraction.

Feat.	Comparison ($A - B$)	$\bar{A}$	$\bar{B}$	95% CI	$\mathcal{C}$	n_0	KS p	MW p
D	Human – Fungal	0.181	−0.125	[−0.178, −0.048] ^z	0.69	2	7×10^{-4}	6×10^{-4}
D	Human – Bacterial	0.143	0.081	[−0.036, 0.250]	0.57	2	0.394	0.853
D	Human – Nematode	0.114	0.010	[−0.123, 0.306]	0.09	5	0.169	0.135
D	Fungal – Bacterial	0.255	0.196	[0.077, 0.328] ^z	0.77	1	3×10^{-3}	0.016
D	Fungal – Nematode	0.195	0.135	[−0.013, 0.376]	0.69	1	6×10^{-3}	3×10^{-3}
D	Bacterial – Nematode	0.174	−0.065	[−0.251, 0.261]	0.37	3	0.112	0.137
V	Human – Fungal	0.114	0.063	[−0.072, 0.280]	0.55	4	0.018	5×10^{-3}
V	Human – Bacterial	0.773	0.684	[0.633, 0.731] ^z	0.88	1	$< 10^{-4}$	$< 10^{-4}$
V	Human – Nematode	0.109	−0.085	[−0.196, 0.285]	0.78	4	0.447	0.502
V	Fungal – Bacterial	0.769	0.690	[0.554, 0.736] ^z	0.90	1	$< 10^{-4}$	$< 10^{-4}$
V	Fungal – Nematode	0.194	−0.153	[−0.311, 0.217]	0.79	3	0.691	0.389
V	Bacterial – Nematode	0.794	−0.701	[−0.743, −0.547] ^z	0.88	1	$< 10^{-4}$	$< 10^{-4}$

V -based indices for every bacterial comparison are large ($\bar{A} \approx 0.77$ – 0.79 , $\mathcal{C} \approx 0.88$ – 0.90) while the corresponding D -based indices are modest, i.e. the proxy manufactures a taxonomic contrast that the experimental annotations do not support.

3.3. Reconciliation with the established performance of pLDDT

The result above must not be read as contradicting the literature that validates pLDDT as a disorder signal [9, 10, 11]. Three distinctions separate the two claims, and each supplies a candidate mechanism.

First, the quantities differ in level. Published performance figures are residue-level discrimination statistics — $F_{\max}$, average precision or AUC over annotated residues [12, 13, 14] — whereas $\rho(D, V)$ is a between-protein rank correlation of aggregate content. Aggregation destroys positional information, and evaluations that do target protein-level disorder content report weaker and more variable performance than residue-level benchmarks [15]. A predictor can label residues well and still order proteins poorly by total fraction.

Second, and most concretely, the threshold is wrong for the task. The very-low-confidence band is defined at pLDDT < 50 [5, 6], whereas the empirically optimal dis-

order cutoff on the CAID DisProt benchmark corresponds to pLDDT < 68.8 [9]. The field V therefore counts only the most extreme tail of low-confidence residues and should be expected to under-report curated disorder systematically. That is exactly the direction of the observed offset in three of four taxa, and it is largest where models are most confident. This is a strong reason to regard the negative result as a property of *this particular precomputed field* rather than of pLDDT.

Third, we cannot attribute the taxon dependence to biology rather than technique. Compact bacterial proteomes with deep alignments plausibly do yield higher-confidence models — AlphaFold accuracy and confidence degrade at shallow alignment depth [5], and pLDDT in disordered regions varies with conservation level [25] — and *E. coli* proteins in this sample are the shortest (median 267 residues), while shorter proteins tend to carry lower disorder content. But differences in AlphaFold model version [7, 22], alignment depth, proteome release and DisProt ascertainment are equally consistent with the observation, and none can be separated with the data retrieved. The result is a documented discrepancy, not an established biological difference.

3.4. Estimator dependence of the scalar indices

Nine specifications per comparison were evaluated: kernel bandwidth scaled to 0.75, 1.0 and 1.25 of Scott’s rule; histograms with 10, 20 and 40 bins; and $\varepsilon \in \{10^{-6}, 10^{-2}, 10^{-1}\}$ (Fig. 5). Two conclusions follow.

The kernel results are stable. Across all bandwidths and all ε values tested, no comparison changed the sign of $\bar{B}$, and magnitudes moved modestly (for human versus fungal on D , $\bar{B}$ ranged over -0.127 to -0.094 , the smallest value arising at $\varepsilon = 10^{-1}$, where regularisation begins to compress the profile).

The histogram results are not. Sign changes occurred in 5 of 12 comparisons, and every one was produced by histogram estimation rather than by bandwidth or ε ; for human versus fungal on D , $\bar{B}$ moves from -0.125 (kernel) to -0.0002 (20 bins) to $+0.035$ (40 bins). The mechanism is visible in $\bar{A}$, which inflates from 0.181 to 0.333 over the same specifications: empty bins in the smaller group leave one density at zero while the other is positive, so $|S| \rightarrow 1$ over those intervals and accumulates spurious magnitude. This is the concrete form of the ε caveat of Section 2.2.1. $\bar{A}$ is therefore not comparable across estimators, and the framework requires a smooth density estimate with an explicitly reported bandwidth. The seven comparisons whose sign is stable under all nine specifications are recorded in `sign_stability.csv`.

3.5. Limitations

Ascertainment. DisProt proteins are included because their disorder has been experimentally investigated [11, 19]. They are not a random sample of any proteome, and the four groups are not matched on length, function, subcellular localisation or study intensity. No proteome-wide statement follows from any number reported here, and no interval reported here is a proteome-wide window of any kind. Because the UniProt API [20] was unreachable, the planned comparison against reference-proteome metadata was not performed, so the magnitude of the ascertainment bias is undocumented rather than small.

Absent residue-level evidence. The central pLDDT calibration question, whether low confidence marks annotated disordered residues comparably across taxa, is untouched by this report. A protein-level discordance is compatible with good residue-level detection (Section 3.3), and the two must not be conflated.

Sample size. With $n = 58$ nematode and $n = 146$ bacterial proteins against $n = 1339$ human, bootstrap intervals for the smaller groups are wide and several are uninformative. Comparisons involving the nematode group support no directional conclusion.

Negative-residue assumption. Eq. (5) treats every residue outside a curated disorder segment as contributing to the denominator. This is not equivalent to experimental evidence of order; DisProt curation is region-targeted,

and unannotated residues are largely uncharacterised. The same asymmetry is recognised in CAID, where benchmark variants differ in how unobserved residues are treated [13].

Unresolved provenance. The definition behind V , the cause of the 103 missing values, and the gap between exported regions and `regions_counter` all remain open pending database access.

4. Conclusion

The bounded asymmetry profile is a usable descriptive instrument: it is well defined, bounded, cheap to compute, and it exposed localised structure in a comparison that global tests reported as null. Its scalar magnitude index, however, is estimator-dependent in a way that must be controlled by specifying a smooth estimator and reporting its bandwidth.

On the gating question we classify the outcome as a *substantial cross-taxon difference in the AlphaFold-derived proxy*. The very-low confidence fraction is weakly concordant with curated disorder content within every taxon examined and carries a taxon-dependent offset that is largest for *E. coli*. Because the very-low band is defined at a threshold well below the empirically optimal disorder cut-off [9], the most parsimonious reading is that this precomputed field is miscalibrated for protein-level disorder content, not that pLDDT lacks disorder signal. Either way, applying it as a taxon-neutral disorder feature would confound calibration with biology.

The next step is consequently not a proteome-scale application. It is to obtain residue-level pLDDT with recorded model versions for these same 1,766 accessions, verify extraction against both mmCIF and the confidence JSON, and test detection of annotated disordered residues separately within each taxon — at the optimal cutoff rather than the very-low band boundary. Should residue-level calibration prove to differ as the protein-level offsets suggest, a common sequence-based disorder predictor applied uniformly across taxa, of the kind assessed in CAID3 [14], is the appropriate substitute for pLDDT in cross-taxon work.

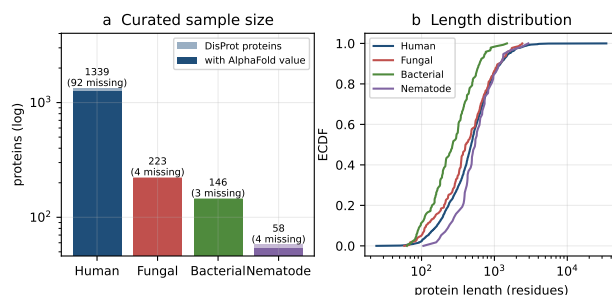


Figure 1. Curated sample composition. (a) Proteins per group with the number lacking an AlphaFold value; note the log scale and the 23-fold range in group size. (b) Protein-length ECDFs.

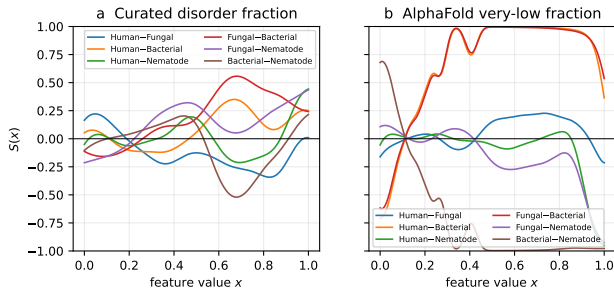


Figure 2. Bounded asymmetry profiles $S(x)$ for all six pairwise comparisons, for (a) curated disorder fraction and (b) the AlphaFold very-low fraction. Kernel estimate, Scott's rule, $\varepsilon = 10^{-3}$. The bacterial comparisons in (b) approach the bound $|S| \rightarrow 1$ at low x ; the corresponding panel-(a) profiles do not.

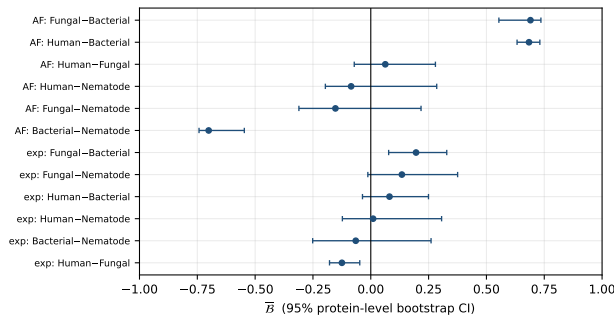


Figure 3. Signed balance index with 95% protein-level bootstrap intervals (2,000 replicates). Only four of twelve comparisons exclude zero; three of those four involve the AlphaFold proxy against *E. coli*.

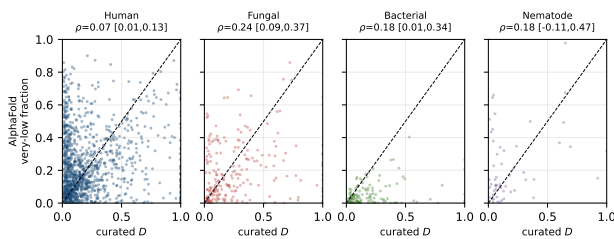


Figure 4. Within-protein concordance between curated disorder fraction and the AlphaFold very-low fraction, with Spearman ρ and bootstrap intervals. Dashed line is identity. Concordance is weak in all four groups; the bacterial panel is compressed toward the axis.

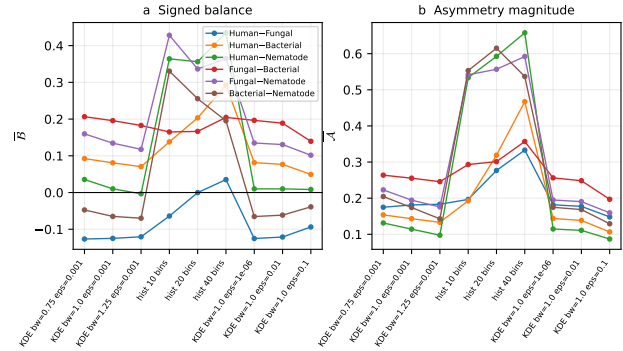


Figure 5. Sensitivity of (a) $\bar{S}$ and (b) $\bar{A}$ across nine estimation specifications, curated disorder fraction. Kernel specifications are mutually consistent; histogram specifications flip signs and inflate $\bar{A}$ with bin count.

Ethical Statement

This study does not involve human participants, animals, or the use of any personal or identifiable data. All analyses were performed on publicly available, previously published database records. No ethical approval or informed consent was required for this work.

Conflicts of Interest

The author declares that he has no conflicts of interest to this work.

Data Availability Statement

All reported values derive from four DisProt release 2026_06 per-organism JSON exports; file names, byte counts and MD5 checksums are recorded in `source_file_audit.csv`. The following derived tables, analysis scripts and figure sources accompany this report:

- `disprot_protein_level.csv`, `disprot_region_level.csv` — Phase 1 tables
- `thales_curated_sample_results.csv` — pairwise indices, bootstrap intervals and global tests
- `cross_taxon_calibration_summary.csv`, `gap_by_taxon_tests.csv` — concordance results
- `robustness_sensitivity.csv`, `sign_stability.csv` — estimator sensitivity
- `alphafold_retrieval_manifest.csv`, `failed_accessions.csv` — retrieval failures

No network retrieval succeeded; all attempts and their failure reasons are logged. Archival deposit: <https://doi.org/10.5281/zenodo.21627908>

Author Contribution Statement

Elias De Jesús: Conceptualization, Methodology, Software, Formal analysis, Data curation, Validation, Visualization, Writing — original draft, Writing — review & editing.

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